Schelling-Point Reputation Communities: A Research Prototype for Agent-to-Agent Escrow and Arbitration

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Abstract

AI agents that can negotiate and transfer funds need identity, escrow, dispute resolution, and accountability mechanisms. We describe AgentTrust, a research prototype that combines an ERC-721 agent registry, guaranteed escrow, staked voting, and an on-chain reputation counter. This paper separates three kinds of evidence: properties of a legacy Solidity snapshot, accounting identities under explicit assumptions, and exploratory Monte-Carlo results. It does not claim production readiness, a security audit, formal verification, public deployment, or a complete implementation of the proposed reputation–guarantee feedback loop. The legacy voting rule uses a 65% threshold, not two thirds; abstention is refunded and can weakly or strictly outperform voting; a minority can cause a void outcome even when it cannot select its preferred verdict; vote deposits are refundable capital at risk rather than an unconditional attack expenditure; and integer division can leave dust. Because contracts and simulations are being revised in parallel, concrete implementation statements are scoped to the inspected legacy snapshot and must be revalidated against a frozen release. The result is an implementable research agenda and testable prototype, not a deployed trust guarantee.

Keywords: AI agents; blockchain; decentralized arbitration; staked voting; reputation; guaranteed escrow; research prototype.

1 Introduction

Large language models and tool-use systems can plan, call external services, and initiate economically consequential actions. Open agent-to-agent commerce therefore raises familiar but important questions about identity, escrow, evidence, dispute resolution, and liability [1, 2]. Cheap pseudonyms complicate accountability, while opaque execution makes the merits of many disputes difficult to verify automatically.

Problem and scope. We study a narrow systems question: what can be learned from coupling principal-bound agent identifiers, collateralized escrow, a simple staked vote, and outcome counters in one prototype? The prototype does not establish the off-chain truth of a dispute, legal identity, or legal liability. It also does not solve secure evidence collection, voter selection, bribery, privacy, oracle failure, or contract governance. “Principal-bound” below means only that a token and stored metadata are associated with an on-chain address; it is not proof that the address corresponds to a unique human or legally responsible entity.

Prototype architecture. The identity–trust–accountability (ITA) decomposition in Section 3 maps to four legacy Solidity contracts: AgentRegistry, GuaranteeEscrow, SchellingVoting, and ReputationHub. The intended research design is a feedback cycle in which outcomes update reputation and reputation informs later guarantee quotes. In the inspected legacy snapshot, however, the contracts record outcomes and accept a caller-supplied premium; they do not compute or enforce a reputation-dependent premium. The closed loop is thus a proposed policy layer whose integration remains to be verified in the parallel implementation.

Version statement. Implementation descriptions in this paper refer to the repository state inspected on August 10, 2026, before the concurrent contract and simulation revisions were frozen. They are descriptive, not a claim about the final code. Historical repository material reports four contracts and 38 Foundry tests, but this paper does not treat that count as an audit or as evidence about untested behaviors. No new implementation result from the in-progress revision is reported here.

Contributions.

- A research-prototype architecture connecting identity records, collateralized escrow, staked dispute voting, and outcome counters.
- A code-scoped description of the legacy 65% rule, including quorum, void-case, abstention-refund, capital-at-risk, and integer-dust behavior.
- Conditional accounting and threshold propositions with explicit hypotheses, replacing unsupported claims of strict Bayesian–Nash truthfulness, harmless minorities, unprofitable Sybils, and exact conservation.
- An exploratory evaluation report that distinguishes simulation observations from contract verification and documents what must be rerun after a release is frozen.

Organization. Section 2 reviews related systems. Section 3 presents the ITA architecture and threat boundaries. Section 4 states the mechanism and limited analytical results. Section 5 describes the legacy implementation and exploratory evaluation. Section 6 concludes with limitations and a validation plan.

2 Related Work

2.1 Identity and credentials

W3C Decentralized Identifiers and Verifiable Credentials provide identifier and claim data models [3, 4]. Non-transferability proposals such as EIP-5192 and EIP-4973 explore account-bound attestations [5, 6], while ERC-4337 supports programmable account logic [7]. These primitives do not by themselves establish unique personhood, legal responsibility, truthful behavior, or fair adjudication. Recent agent-identity proposals emphasize zero-trust controls and verifiable bindings [8, 9]; AgentTrust uses a much weaker on-chain address binding in the current prototype.

2.2 Reputation, attestations, and insured agents

Ethereum Attestation Service supplies general attestation infrastructure [10], and recent work studies blockchain agent reputation and registered-agent datasets [2, 11]. Insured Agents proposes a

Table 1: ITA layers and current evidence boundary.

Layer	Prototype component	What is and is not established
Identity	AgentRegistry	Associates an ERC-721 token and metadata with an address. It does not verify a unique human, organization, beneficial owner, legal capacity, endpoint integrity, or non-transferability.
Trust	ReputationHub	Counts selected outcomes from allowlisted callers and computes a heuristic score. It does not authenticate evidence, prevent collusive history creation, or enforce a premium derived from the score.
Account.	GuaranteeEscrow and SchellingVoting	Implements custody, timeouts, a single-round vote, and payouts in the inspected snapshot. It does not establish factual correctness, neutral governance, juror independence, appeal, or legal enforceability.

market in which specialized insurers post collateral and price agent risk [12]. Our prototype is related, but its scalar score is only a heuristic outcome summary. The inspected contracts do not estimate default probability, quote premiums, or prevent wash trading and score farming. Any reputation-to-price equation in Section 4.6 is a modeling hypothesis, not implemented underwriting.

2.3 Decentralized dispute resolution

Kleros, Aragon Court, and UMA use staking, challenge, and/or coordination incentives in deployed or documented dispute systems [13, 14, 15]. Nexus Mutual illustrates capital pooling and claims assessment [16]. These systems show that stake can make identities and actions costly, but they do not make truth a dominant strategy: bribery, voter selection, information quality, participation, governance, and correlated failures remain central. AgentTrust currently implements a single-round open vote without random jury selection or an appeal layer.

2.4 Agent trust infrastructure

Recent work surveys blockchain-based trustworthy agent networks and agent-to-agent finance [17, 1]. ERC-8004 specifies interoperable agent registries [18]. These works motivate combining identity, payment, verification, and reputation. We make no priority or "first system" claim. The relevant design space is active, and local prior work already includes closely related identity, insurance, escrow, and governance components. AgentTrust should therefore be read as one concrete composition and experimental test bed rather than as a uniquely complete or production-proven protocol.

3 The ITA Architecture

We use identity–trust–accountability (ITA) as a decomposition, not as a claim that all three properties are achieved. Table 1 separates the intended role from what the legacy prototype actually establishes.

3.1 Threat model

The modeled adversary may create addresses, coordinate voters, bribe or side-pay participants, selectively abstain, trade with related parties, and exploit public transaction ordering. Traders may file frivolous disputes; guarantors may be undercapitalized outside a single trade; and privileged owners may configure writers or case creation. We assume standard cryptographic primitives and the underlying chain execute as specified, but we do not assume that an on-chain address maps one-to-one to a person or that a vote signal is independent and accurate.

The following threats are outside the current mechanism and therefore remain limitations rather than solved properties: compromised private keys; malicious or upgradeable dependencies; front-running; denial of service; evidence withholding; coercion; bribery markets; governance-key compromise; contract insolvency caused by forced Ether or unrelated balances; token-price volatility; chain reorganizations; privacy leakage; regulatory treatment; and off-chain enforcement. The legacy prototype has not been independently audited or formally verified.

3.2 Trust boundaries and privileged roles

The registry owner can set and withdraw registration fees. The reputation owner chooses authorized writers. Case opening is owner-gated in the inspected voting contract. The escrow's owner can resolve disputes; a deployment script may transfer that ownership to the voting contract, but the paper has not independently verified a public deployment or every deployment path. These privileges mean the prototype is not fully decentralized. A release claim would require a frozen deployment manifest, verified bytecode, owner and ACL checks, and tests of the complete linkage.

3.3 Intended feedback loop

The proposed loop is

$$\text{outcome} \rightarrow \text{reputation record} \rightarrow \text{risk estimate} \rightarrow \text{guarantee quote} \rightarrow \text{future trade.}$$

Only the first arrow and score computation are directly represented in the inspected contracts. A market or policy engine must implement the remaining arrows, including calibration, freshness, minimum data, uncertainty, and resistance to manipulated histories. Until that code is frozen and verified end to end, we call this an *open-loop prototype with a proposed closed-loop design.*

4 Mechanism Design

This section describes the inspected legacy mechanism and states only code-level or conditional results. It deliberately does not assert that truthful voting is a strict Bayesian–Nash equilibrium, that a 35% coalition is harmless, that a Sybil attack is unprofitable, or that all funds are exactly conserved. Those conclusions require assumptions or implementation properties absent from the current evidence.

4.1 Legacy mechanism

For one disputed trade, let $m_B, m_S, m_\perp$ be the counts of BUYER, SELLER, and ABSTAIN votes. Each address that votes deposits a fixed amount $v > 0$. Define the effective-vote count $T = m_B + m_S$; abstentions are excluded from both the threshold denominator and the quorum. The legacy constants are

$$\kappa = \frac{6500}{10000} = 0.65, \qquad \nu = 3.$$

Thus the mechanism does *not* implement an exact two-thirds rule.

Definition 4.1 (Legacy adjudication rule)**.** *If $T = 0$, the case is void. Otherwise side $X \in \{B, S\}$ is selected when $m_X \cdot 10000 \geq T \cdot 6500$. A selected side produces an effective verdict only when $T \geq \nu$; otherwise the case is void. Since $0.65 > 1/2$, both sides cannot satisfy the predicate when $T > 0$. In the inspected snapshot a void case triggers a full buyer-wins escrow settlement, while all vote deposits are claimable as refunds.*

Definition 4.2 (Legacy voter settlement)**.** *In an effective case, each voter on the selected side receives*

$$v + \left\lfloor \frac{v m_\ell}{m_w} \right\rfloor,$$

where m_w and m_ℓ are the winning and losing effective-vote counts. Losing voters receive zero, and abstainers may reclaim v. In a void case every participant may reclaim v. Claims are pull payments and therefore require each eligible address to transact successfully.

Escrow linkage. The legacy escrow holds buyer amount x and guarantor collateral

$$g = \left\lfloor \frac{x\gamma}{10^{18}} \right\rfloor.$$

The premium π is quoted but is not separately deposited. The guard is $\pi \leq x$, so `premium == amount` is accepted and makes the seller payment zero on a seller-win or normal-release path; the code does not require a positive seller remainder. On a full buyer win, the buyer is sent $x + g$. On a seller win or normal release, the seller is sent $x - \pi$ and the guarantor is sent $g + \pi$. Partial buyer resolution sends the buyer $\lfloor x\beta/10000 \rfloor + g$ and the seller the remaining buyer principal.

4.2 What the threshold does and does not imply

Proposition 4.3 (Vote count required to select a side)**.** *Suppose h effective votes oppose a coalition and all c coalition votes support one side. That side is selected by the legacy rule if and only if*

$$c \geq \left\lceil \frac{13h}{7} \right\rceil.$$

This is a deterministic statement about a fixed vote profile, not a profitability or truthfulness claim.

Proof. The side is selected exactly when $c/(c + h) \geq 0.65 = 13/20$. Rearranging gives $7c \geq 13h$; integrality gives the ceiling. The separate quorum condition must also hold. $\square$

A coalition with at most 35% of effective votes cannot by itself select its preferred side against unanimous opposition. It is nevertheless not "strategically harmless." If the opposing side has less than 65%, neither side is selected and the case is void. Because void cases default to the buyer, a coalition can change the economic outcome by inducing a void, particularly when it favors the buyer. It can also bribe, split honest votes, exploit abstention, or benefit from signal errors. The threshold provides a vote-count barrier, not general collusion resistance.

4.3 Abstention and equilibrium status

The legacy payoff rule creates an unresolved participation problem. An abstainer receives net monetary payoff zero after refund. A voter receives a positive bonus only if on an effective winning side, loses v if on an effective losing side, and receives zero net payoff if the case is void. If p_w and p_ℓ denote the probabilities that a contemplated vote is respectively winning and losing, its expected net payoff is

$$p_w \left\lfloor \frac{vm_\ell}{m_w} \right\rfloor - p_\ell v,$$

with the counts outcome-dependent. Voting beats abstention only when this expression is positive. Signal accuracy $q > 1/2$ alone does not establish that inequality, because the threshold, pivotality, other strategies, correlated signals, and the possibility of a void all matter.

Consequently, the earlier strict-Bayesian–Nash statement and its proof are withdrawn. A future analysis could establish a *conditional* equilibrium under a fully specified prior, evidence model, electorate-size distribution, participation reward, and strategy profile. The current score contract records seller trade outcomes, not voter participation, so the proposed reputation loop does not presently turn abstention into a repeated-game penalty. That integration remains future work.

4.4 Sybil cost is capital at risk, not proven loss

One address may vote once per case, but the inspected voting contract does not require the address to hold a registered agent token. Creating additional voting addresses is therefore cheap. Each vote must lock v until settlement and claim, which creates a linear *capital requirement*. It is not an equal linear expenditure: a winning Sybil recovers principal and may earn a reward; a void-case Sybil can reclaim principal; only a losing effective vote forfeits principal. Registration fees are irrelevant to the legacy voting path unless a future release binds voters to registered identities. Even with such a binding, whether a fee is a social cost, protocol revenue, refundable capital, or attacker-controlled transfer depends on governance and ownership.

No "Sybil attacks are unprofitable" theorem follows from Proposition 4.3. Profit also depends on the dispute value, side payments, opportunity cost and duration of locked capital, probability of each outcome, gas, registration policy, and whether the attacker benefits from a void. The prototype establishes only that each legacy vote requires contemporaneous stake.

4.5 Conditional accounting

Proposition 4.4 (Voting-pool claim accounting)**.** *Assume a case reaches settlement; every eligible claimant claims exactly once; every transfer succeeds; and no unrelated Ether is included. Then a void case makes claims totaling* $(m_B + m_S + m_\perp)v$*. An effective case makes claims totaling*

$$(m_B + m_S + m_\perp)v - \delta, \quad \delta = (vm_\ell) \bmod m_w, \quad 0 \le \delta < m_w.$$

The remainder δ *remains in the contract unless a separate recovery mechanism exists.*

Proof. The winning side claims $m_w v + m_w \lfloor vm_\ell/m_w \rfloor = m_w v + vm_\ell - \delta$; abstainers claim $m_\perp v$ and losers claim zero. The void expression follows because every participant is eligible for principal. These are claim entitlements, not a liveness guarantee. □

Proposition 4.5 (Escrow branch accounting)**.** *For the legacy payout formulas, if the contract enters the stated branch, all calls succeed, and its trade-specific custody is exactly* $x + g$*, the nominal payout sums are* $x + g$ *on full buyer win, seller win, normal release, and partial buyer resolution.*

Before guarantee, the funded-timeout path pays x. This proposition does not prove that every reachable state has a live exit or that the contract balance always equals the sum of trade-specific liabilities.

Proof. The sums are respectively $(x + g)$, $(x - \pi) + (g + \pi)$, the same normal-release sum, and $\lfloor x\beta/10000 \rfloor + g + x - \lfloor x\beta/10000 \rfloor$. The identities permit $\pi = x$ and hence a zero seller payment. $\square$

These propositions replace "exact conservation." They do not cover unclaimed balances, failed recipient calls, forced Ether, cross-trade balance commingling, dust recovery, liveness, or future code. Moreover, the legacy escrow performs external value transfers before setting the terminal state in internal release helpers. Entrypoints use `nonReentrant`, but that ordering is not the canonical checks–effects–interactions pattern and should not be described as such.

4.6 Reputation and guarantee pricing hypothesis

The legacy reputation score is a descriptive counter-derived quantity. If $\rho = (c, d, w, l)$ and $N = c + d + w + l > 0$, the inspected code computes, with Solidity integer arithmetic,

$$s(\rho) = 100 - \left\lfloor \frac{100d}{N} \right\rfloor - \left\lfloor \frac{50l}{N} \right\rfloor,$$

and returns 50 when $N = 0$. The score is not a calibrated probability and ignores transaction value, recency, domain, buyer behavior, uncertainty, and correlated counterparties.

For illustration only, if an external underwriter independently estimates a default probability $p_D \in [0, 1)$ and loses collateral g on default while earning premium π otherwise, a zero-profit condition under risk neutrality is

$$(1 - p_D)\pi - p_D g \geq 0 \quad \iff \quad \pi \geq \frac{p_D}{1 - p_D} g.$$

This is a conditional actuarial identity, not an on-chain rule. Substituting $p_D = (100 - s)/100$ would be an unvalidated modeling choice and can imply a quote above the legacy constraint $\pi \leq x$. The current contracts neither make that substitution nor enforce the floor. Closing the loop requires empirical calibration, uncertainty bounds, manipulation resistance, and final end-to-end code verification.

4.7 Threats to validity

The analysis assumes a binary dispute despite ambiguous or partial performance; ignores evidence quality and correlated signals; treats stake utility as linear; and omits bribery, hedging, token volatility, gas, latency, and identity resets. Open voting reveals interim information and permits late strategic voting. A fixed three-vote quorum is weak for high-value trades. Buyer-favoring void settlement creates a filing and nonparticipation attack surface. Registration tokens are transferable under ordinary ERC-721 behavior in the inspected registry, so "soulbound" or permanent principal binding is not established. These limitations prevent security or welfare conclusions beyond the narrow propositions above.

5 Prototype Implementation and Exploratory Evaluation

Scope. This section describes the legacy repository snapshot inspected on August 10, 2026. Contracts and simulations are being revised in parallel, so the statements below must not be read as results for the forthcoming version. A release paper should identify a commit, compiler settings, dependency lock, deployment addresses, chain, test command, and artifact hashes. None is claimed here as a finalized release manifest.

5.1 Legacy contract snapshot

The snapshot contains four Solidity `0.8.24` contracts using OpenZeppelin components:

- AgentRegistry mints transferable ERC-721 tokens, stores descriptive metadata and an address in `AgentInfo.owner`, and accepts an owner-configurable registration fee. This is an address association, not proof of personhood, legal identity, or non-transferability.
- GuaranteeEscrow implements the states `CREATED`, `FUNDED`, `GUARANTEED`, `DELIVERED`, `DISPUTED`, `RELEASED`, and `RESOLVED`. It has one-day funding, guarantee, delivery, and confirmation windows, but the inspected `dispute` call has no separate deadline.
- SchellingVoting implements open, one-address-one-vote cases with a fixed stake, 65% threshold, three-effective-vote quorum, refunded abstention, and pull-based claims. Case creation is owner-only. Voters are not required to hold registry tokens.
- ReputationHub stores four seller-side counters and permits writes from addresses allowlisted by its owner. The voting contract need not be a writer because the escrow records the seller's dispute outcome. The ACL prevents arbitrary direct writes only if governance configures and protects it correctly.

The escrow's `resolveDispute` is owner-only. The inspected deployment script intends to transfer escrow ownership to the voting contract and to authorize writers, but this paper does not claim that a public deployment with those links has been verified. The owner-only case-opening path also remains a privileged control point.

Premium and payout semantics. The guarantee call enforces `premium <= amount`; equality is valid. The premium is accounting metadata netted from buyer-funded principal, not additional custody. Therefore the seller receives `amount - premium`, which can be zero, and the guarantor receives collateral plus premium on a normal release or seller win. Descriptions that said equality was rejected or that the seller share was necessarily positive were incorrect and have been removed.

External-call ordering. The internal escrow helpers call recipients before writing the terminal state. Public entrypoints are protected by `nonReentrant`, which mitigates same-contract reentry, but transfer-before-state is not canonical checks–effects–interactions. A failed recipient call reverts the transaction; a recipient that continually rejects Ether can therefore impair liveness. The prototype has not been audited for cross-contract callbacks, forced Ether, griefing, or all recipient behaviors.

5.2 Testing evidence and its limits

Historical repository documentation and test files enumerate 38 Foundry tests: eight registry tests, ten escrow tests, eleven voting tests, eight reputation tests, and one end-to-end scenario. That is a useful regression suite, but this paper did not rerun or attribute the count to the concurrently changing contracts. Passing unit tests would not constitute formal verification, an audit, or proof of production safety. The existing suite is example-based and does not by itself establish:

Table 2: Selected legacy simulation observations. Values are descriptive outputs from existing local result files and require rerunning against a frozen implementation.

Experiment	Observation and appropriate interpretation
Voting strategy	With 15 voters and 3000 repetitions per signal-accuracy setting, the reported honest deviator mean payoff ranges from +0.005 to −0.011 ETH, while abstention is exactly zero. This exposes the abstention problem; it does not establish strict truthfulness.
Collusion	With 21 voters and $q = 0.85$, the reported seller-flip rate rises from near zero at small coalitions to 0.397 at 12 colluders and 1 at 14. A non-flip may instead be a buyer-favoring void, so flip rate alone does not measure strategic harmlessness.
Sybil sweep	The modeled total labeled "cost" equals number of identities times registration fee plus vote stake. Because vote stake is generally refundable for winners and void cases, the quantity is better interpreted as assumed fee plus capital locked or at risk, not realized attack loss.
Accounting	Four thousand modeled cases report zero deviation from the simulation's expected claim-accounting formula. This checks the simulator's arithmetic, including its dust term; it is not a whole-contract conservation proof.
Threshold scan	At $n = 15$ and $q = 0.7$, the reported 6500-BPS run has correct-verdict rate 0.721 and void rate 0.275. This illustrates a threshold/void tradeoff under one parameterization, not an optimality result.

- exhaustive state-machine reachability and liveness;
- aggregate liabilities across concurrent trades and cases;
- correctness under arbitrary recipient contracts and forced Ether;
- bribery, Sybil profitability, identity uniqueness, or evidence truth;
- deployment linkage, governance-key security, or bytecode/source correspondence.

Before making implementation claims, the frozen release should run unit, fuzz, invariant, and integration tests; static analysis; a dependency and compiler review; and an independent security audit. Useful invariants include solvency per trade, no double claim, terminal-state reachability, case-to-trade identity consistency, bounded dust with an explicit recovery policy, and ACL/ownership correctness after deployment.

5.3 Exploratory legacy simulation

The repository contains legacy simulation result files generated with a fixed seed. Because the simulation is also being revised, we report them only as exploratory observations about the modeled rule, not as verification of current contracts or proof of equilibrium.

The simulation assumes exogenous strategies, signal accuracy, and electorate sizes. It does not model strategic entry, correlated information, bribery, adaptive voting, gas, latency, claim failure, reputation manipulation, guarantee-market pricing, or implementation-level execution. Reusing the same settlement formula in code and simulation is a consistency check only after an independent trace-level differential test demonstrates correspondence. That differential check remains pending for the parallel implementation.

5.4 Deployment status

The repository includes local Anvil and Base Sepolia deployment instructions. We found no basis in this paper task to claim an independently verified public deployment, verified source, audit, or live production use. Accordingly, "deployed" refers only to local script intent where unavoidable, and not to a public network. A future testnet report should record transaction hashes, chain ID, contract addresses, verified bytecode, owners, authorized callers, configuration values, and an end-to-end transaction trace. Mainnet or real-value operation would additionally require security review, monitoring, incident response, governance, economic stress testing, and legal analysis.

5.5 Current limitations and validation checklist

The central limitations are: free abstention; buyer-favoring void outcomes; a low fixed quorum; open non-commit-reveal voting; no registered-voter requirement; transferable identity tokens; owner-gated case creation and configuration; no appeal; no authenticated evidence format; no calibrated pricing engine; no on-chain enforcement of reputation-based premiums; integer dust; pull-claim liveness; transfer-before-state helper ordering; and an unverified contract/simulation correspondence while both are changing.

A defensible next version should freeze code first, then (1) generate a machine-readable mechanism specification; (2) differentially test every threshold, quorum, payout, and void branch; (3) rerun all experiments from clean artifacts with confidence intervals and raw-data hashes; (4) analyze participation with abstention explicitly; (5) report capital locked, capital at risk, and realized loss as distinct quantities; and (6) state any theorem only with assumptions that match both code and experiment.

6 Conclusion

AgentTrust is a research prototype for studying how agent identifiers, collateralized escrow, staked dispute voting, and outcome counters might interact. The inspected legacy snapshot provides concrete state machines and payout rules, but it does not justify claims of production readiness, audit, formal verification, public deployment, strict truthful equilibrium, general collusion resistance, unprofitable Sybil attacks, or exact conservation.

The corrections are substantive. The implemented threshold is 65%, not two thirds. A coalition that cannot select its preferred verdict may still induce a void, and void defaults to the buyer. Abstention returns principal and has zero net monetary payoff, while honest voting can have negative expected payoff; voter participation is not currently recorded in reputation. Vote stakes are refundable capital at risk except when forfeited, not automatically consumed attack expenditure. Voting claims can leave integer dust, and accounting identities require settlement, successful calls, and complete claims. The premium may equal the trade amount, leaving a zero seller payment. Finally, the proposed reputation–pricing loop is not yet enforced by the inspected contracts.

Future work should begin with a frozen, versioned release and differential contract/simulation tests. Only then should the project revisit equilibrium conditions, calibrated underwriting, voter identity, commit–reveal or jury sampling, appeal, evidence authentication, dust recovery, payout liveness, governance, and adversarial testing. The useful result at this stage is not a broad security guarantee but a transparent prototype, a set of conditional accounting and threshold facts, and a concrete list of hypotheses that can be falsified by code, simulation, and independent review.

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